Muhammad Uzair Warraich

Final-Year BS Computer Science Student, AI, LLM Engineering and Backend Development

Rawalpindi, Pakistan | +92 318 9442827 | uzair2145819@gmail.com | linkedin.com/in/uzyr-chaudhary | github.com/Uzayr-Ch

PROFILE

I am a final-year Computer Science student at PIEAS, Islamabad, specializing in artificial intelligence, large language models, and applied backend engineering. Over two internships I have built a production-style retrieval-augmented generation pipeline, fine-tuned a multilingual LLM, and shipped full-stack AI systems using FastAPI, React, and vector databases. Alongside that, I have a solid portfolio of independent projects in computer vision, deep learning, and data engineering. I like working across the whole stack, from how a model actually behaves and the data feeding it, through to the API and the dashboard someone ends up looking at.

EDUCATION

BS in Computer and Information Sciences *Sep 2023 – May 2027 (Expected)*

Pakistan Institute of Engineering and Applied Sciences (PIEAS), Islamabad. Final Year. CGPA: 3.62 / 4.00

Relevant coursework: Artificial Intelligence, Machine Learning Fundamentals, Computer Vision, Deep Learning Foundations, Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming, Parallel and Distributed Programming (CUDA, OpenMP, Pthreads), Probability and Statistics, Linear Algebra for Computing.

FSc Pre-Engineering *2021 – 2023*

Punjab Group of Colleges. Federal Board of Intermediate and Secondary Education (FBISE). 991 / 1100

PROFESSIONAL EXPERIENCE

AI Backend Intern, AIgilityX Arabia LLC FZ *Jul 2026 – Aug 2026 (8 weeks)*

Hybrid. AI strategy and applied AI solutions company

- Sourced, collected, and cleaned datasets for a multilingual large language model fine-tuning pipeline, handling preprocessing and quality checks before training.
- Contributed to fine-tuning a multilingual Mistral-based LLM as part of the company's model customization work.
- Built backend control panels and internal dashboards on Supabase to support data management and monitoring.
- Worked within an AI and backend development team on data pipeline and application-layer tasks.

AI Intern, NASTP Alpha, Software Development Division (SDH Alpha) *6-week internship*

On-site, Rawalpindi. National Aerospace Science and Technology Park

Project: Automated QMS Engine, an ISO 9001:2015 documentation framework that generates audit-ready quality management documents in under 60 seconds using a controlled retrieval-augmented generation pipeline.

- Designed and built a RAG pipeline that grounds every generated clause in both the organization's real operational data and the exact wording of the ISO 9001:2015 standard, so the output stays specific rather than generic or hallucinated.
- Implemented semantic clause retrieval with ChromaDB over embedded ISO 9001:2015 clause text, plus a rule engine that computes the 29 mandatory and 5 conditional clause applicability matrix deterministically in Python, never through the LLM.
- Applied a fact and prose separation approach: Python code renders every auditable fact (document numbers, dates, signature blocks, clause matrices), while a locally hosted, quantized Llama-3 model running through Ollama, with Groq Cloud used for testing, drafts constrained narrative prose only.
- Built the backend with FastAPI (Python 3.11/3.12, 11 async REST routers, SQLAlchemy ORM, SQLite/PostgreSQL) and a React 18, Vite, and TailwindCSS frontend, including a side-by-side diff review screen for human approval.
- Added a two-layer quality check, combining deterministic Python rule checks with a temperature-0.1 LLM semantic audit, backed by a self-critique retry loop of up to 3 attempts before any draft reaches a human reviewer.
- Enforced a strict 5-stage workflow (intake, drafting, baselined, locked, active) and multi-tenant data isolation, with a per-request tenant guard that returns HTTP 403 on any cross-tenant access, to keep the system production-safe.
- Reached 100% ISO 9001:2015 clause coverage across Chapters 4 to 10, with zero external API dependency in the production inference path and one-click PDF/Markdown export for audit review.

ACADEMIC AND INDEPENDENT PROJECTS

Content-Based Image Retrieval (CBIR) *Independent Project*

Python, PyTorch, OpenCV, FAISS

- Built a high-performance image retrieval system using ResNet50 and Vision Transformers (ViT) for feature extraction and large-scale similarity search.
- Reduced search latency by 40% using FAISS indexing across large image datasets.
- Applied cosine similarity and Euclidean distance metrics to improve retrieval precision across diverse image categories.

Factory Compliance and AI Safety Monitoring System *Independent Project*

Python, YOLOv8, OpenCV, Streamlit, SQLite

- Developed an AI-powered industrial safety compliance system that detects workplace safety violations from surveillance video using YOLOv8 and OpenCV.
- Built a context-aware severity classification and automated escalation pipeline that generates real-time alerts with cryptographically verifiable audit logs.
- Designed an interactive Streamlit dashboard for live monitoring, analytics, historical event tracking, and compliance reporting.

Adaptive E-Learning System *Independent Project*

Python, LLMs, BDI Architecture

- Built an AI-driven adaptive learning platform using 6 autonomous agents and a Belief-Desire-Intention (BDI) architecture for personalized learning.
- Applied Bayesian Knowledge Tracing (BKT) and LLMs to model student learning progress and adjust content delivery.
- Added forgetting-curve algorithms that improved long-term knowledge retention by 25% through smarter review scheduling.

Crop Disease Detection *Independent Project*

Deep Learning, VGG19, Image Processing

- Trained a VGG19-based crop disease classification model on more than 10,000 images, reaching 94% accuracy through data augmentation and preprocessing.
- Improved model generalization across multiple crop species using more advanced image preprocessing techniques.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL

AI and Machine Learning: deep learning, computer vision, neural networks, LLM fine-tuning, retrieval-augmented generation (RAG), prompt engineering, model evaluation, feature engineering

Libraries and Frameworks: PyTorch, OpenCV, Scikit-learn, NumPy, Pandas, FAISS

Backend and APIs: FastAPI, SQLAlchemy ORM, REST API design, async and SSE streaming

Frontend: React 18, Vite, TailwindCSS

Data and Vector Stores: SQLite, PostgreSQL, ChromaDB, Supabase

LLM Infrastructure: Ollama (local inference), Groq Cloud API, Mistral fine-tuning, vector embeddings and semantic search

Tools and Platforms: Git, GitHub, VS Code, Visual Studio, SQL Server, MS Office

Other Strengths: problem solving, technical documentation, cross-functional collaboration, adaptability, independent research

CERTIFICATIONS AND PROFESSIONAL DEVELOPMENT

- Google AI Professional Certificate, Coursera / Google (covers AI fundamentals, brainstorming and planning, research and insights, and writing and communicating with AI)
- Prompt Engineering with OpenAI, Cognitive Class AI
- Working with the OpenAI API, DataCamp
- Understanding Machine Learning, DataCamp
- Understanding AI, DataCamp
- Intermediate Python, DataCamp
- Intermediate SQL, DataCamp
- Data Manipulation with pandas, DataCamp

LANGUAGES

English: professional working proficiency

Urdu: native